API Policies and Best Practices

Cisco has established an API fair use policy to ensure an optimal experience for all Control Center users. Any client-side code that uses Control Center APIs must adhere to the limitations described below.

Concurrent Connections

Cisco advises enterprises to avoid concurrent processing because it has the potential to slow down response time considerably. We recommend using APIs in single-threaded applications or routines only. Accounts using APIs in multi-threaded scenarios may experience throttling or other restrictions, depending on the circumstances.

Policy: Use single-threaded code—one API call at a time.

API TPS

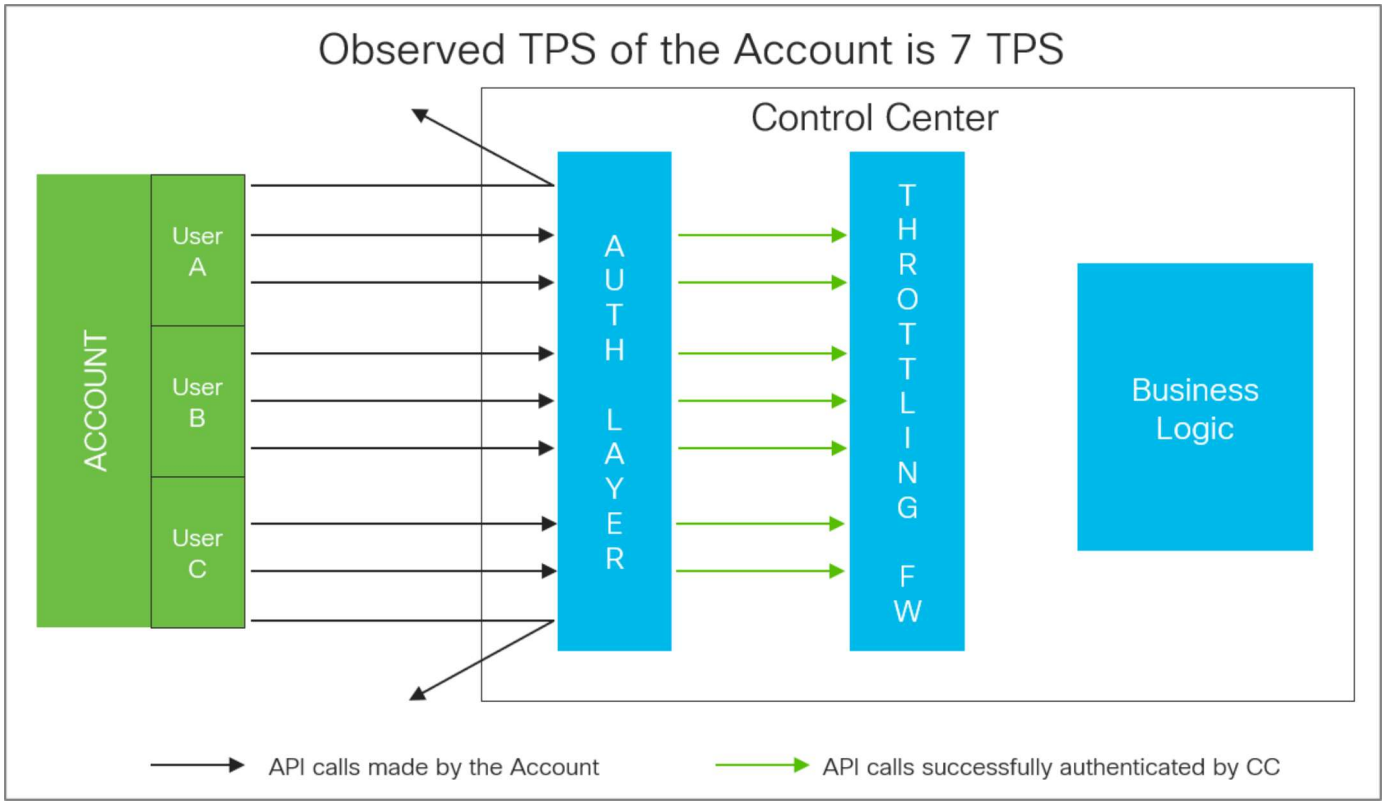
Cisco ensures effective API performance by tracking the number of API calls/requests an enterprise makes per second. This metric is knows as TPS (Transactions Per Second).

When an enterprise exceeds a certain TPS limit, Control Center throttles (rejects) their API requests. Service providers do not have a TPS limit.

Control Center only counts succesfully authenticated API requests toward the enterprise TPS limit. Likewise, Control Center only considers successfully authenticated APIs when throttling for TPS limits. However, Control Center will reject APIs before authentication because of incorrect credentials.

Example

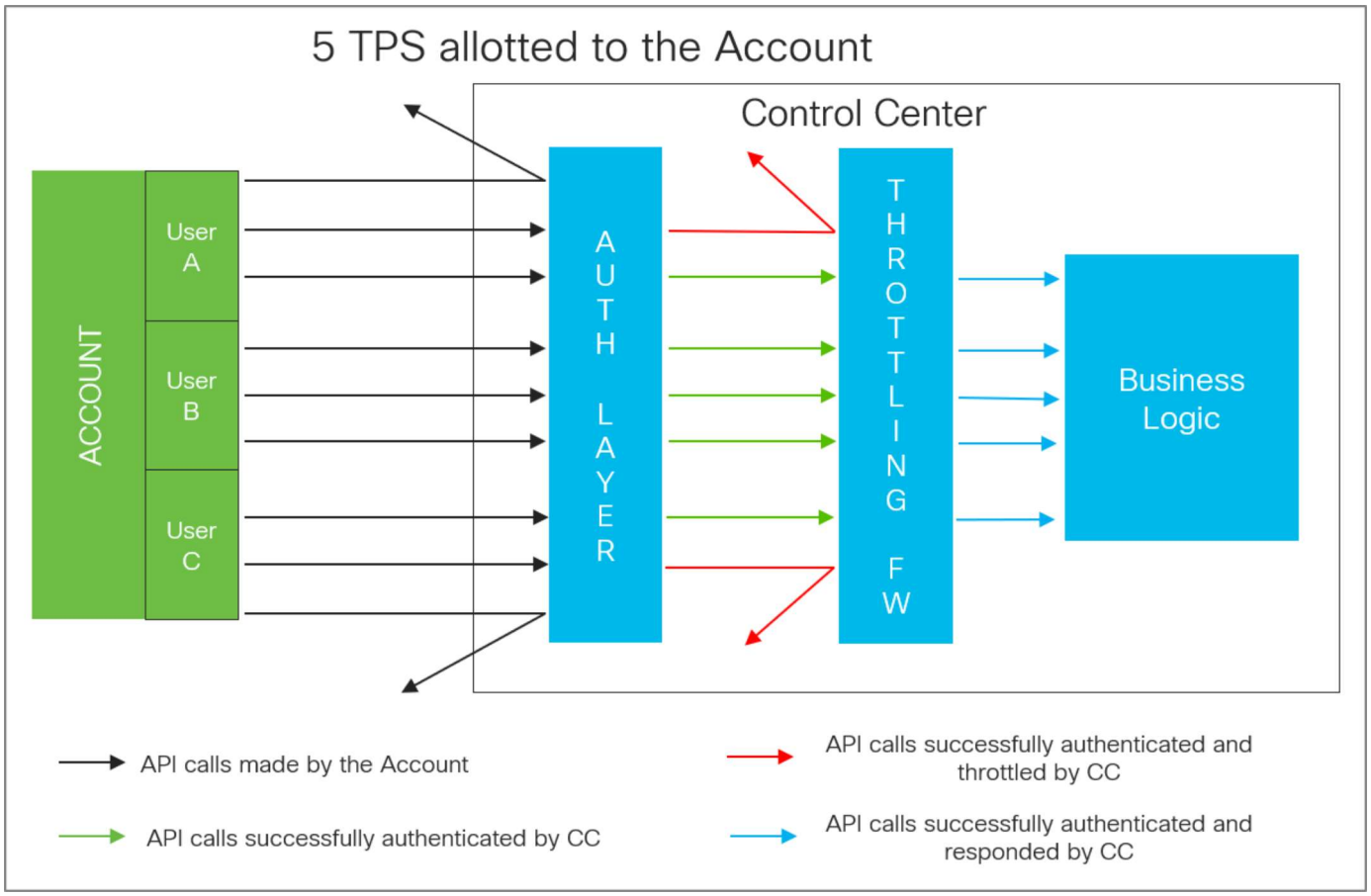
An enterprise has three users. Each user makes 3 API calls in the same second, resulting in a total of 9 API calls for that second. Control Center successfully authenticates 7 of the API calls and rejects 2 of the API calls due to incorrect credentials. Control Center calculates the usage for that second as 7 TPS.



Example

An enterprise with three users has an allotment of 5 TPS. Each user makes 3 API calls in the same second, resulting in a total of 9 API calls for that second. Control Center successfully authenticates 7 API

calls and rejects 2 API calls due to incorrect credentials. The TPS limit for the account is 5, so Control Center accepts 5 of the successfully authenticated API calls and throttles 2 of them.



Base Contracted TPS

Control Center provides a standard amount of API TPS to enterprises. This TPS is known as Base Contracted TPS (BCTPS). The amount an enterprise receives depends on the account type.

Account Type	Base Contracted TPS
Advantage	5 TPS
Essential	1 TPS
Lite	1 TPS

TPS Autoscaling

When an enterprise reaches 1 million devices, Control Center automatically increases the Base Contracted TPS by the Base Contracted TPS amount for no extra charge. Each subsequent addition of

500,000 devices results in another increase. Similarly, each reduction of 500,00 devices results in a decrease of the Base Contracted TPS. This adjusted TPS is known as Scaled-Base Contracted TPS (SBCTPS).

Autoscaling Activation. To activate this feature, the service provider must request the API TPS Service add-on from Cisco and implement Base Contracted TPS for all of their enterprises.

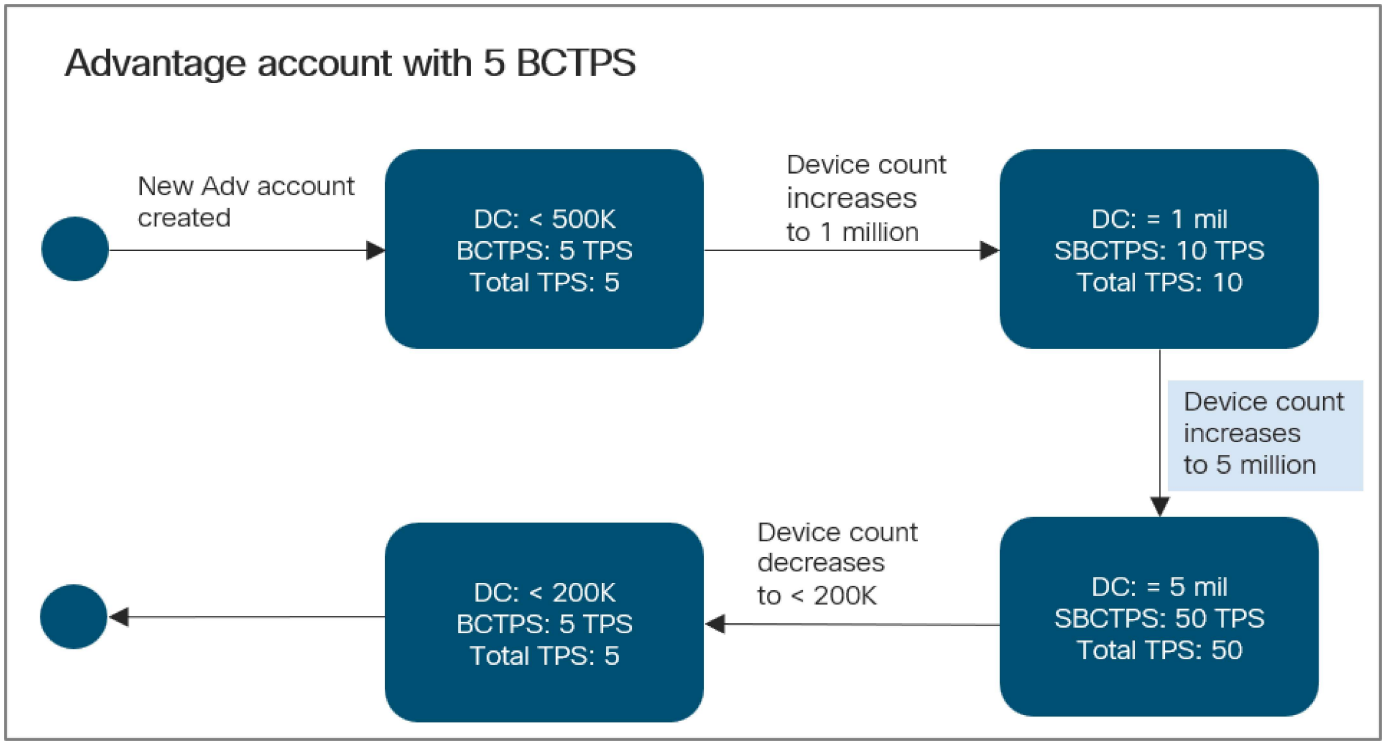
The table below shows the Scaled-Base Contracted TPS for each account type. Control Center increases or decreases the enterprise Base Contracted TPS by this amount when the device count for the enterprise increases or decreases by 500,000, respectively.

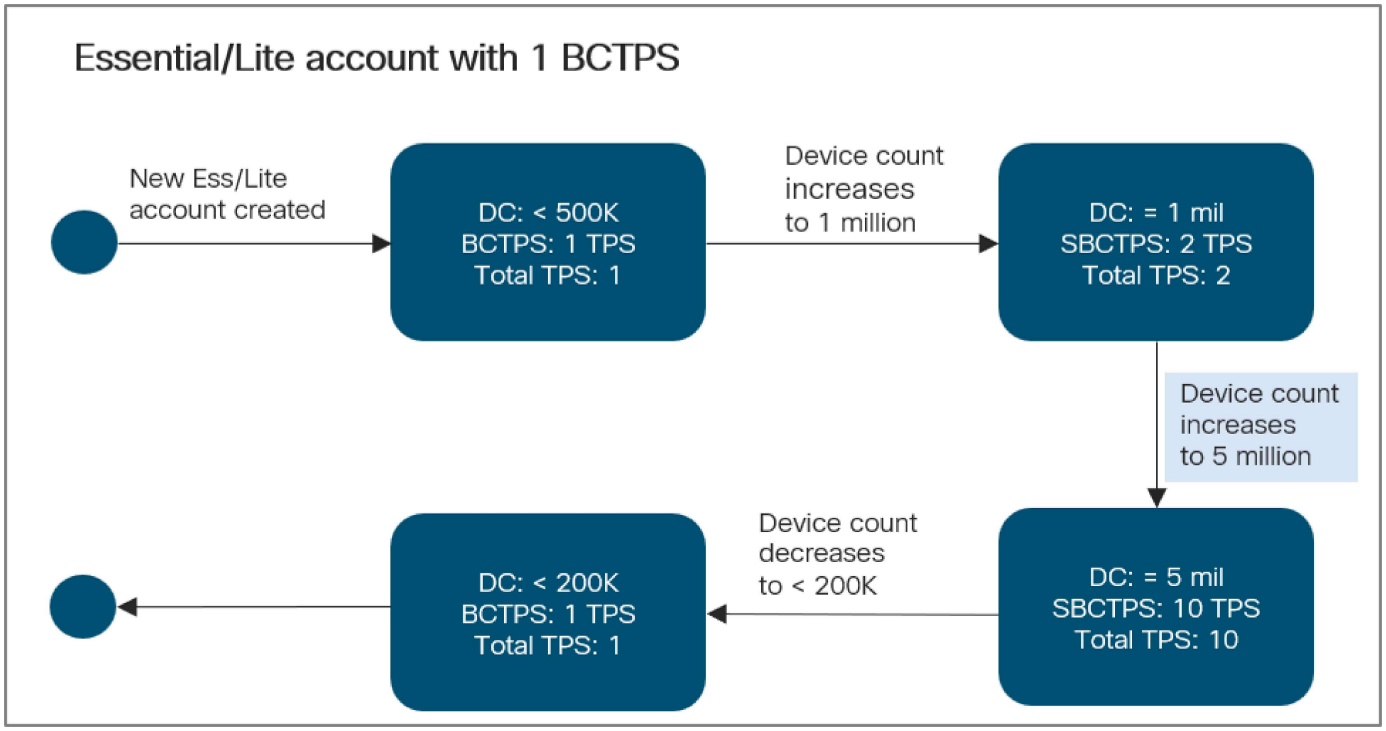
Account Type	**Base Contracted TPS**	**Scaled-Base Contracted TPS**
Advantage	5 TPS	5 TPS
Essential	1 TPS	1 TPS
Lite	1 TPS	1 TPS

Example

An enterprise increases their device count (DC) from 1 million devices to 5 million devices (10 blocks of 500,000 devices).

- For an Advantage account, the new SBCTPS is 50 (10 device blocks * 5 BCTPS).
- For an Essential and Lite account, the new SBCTPS is 10 (10 device blocks * 1 BCTPS).





Example

An enterprise increases their device count from 1 million devices to 2.7 million devices (5.4 blocks of 500,000 devices). In such scenarios, Control Center always rounds the number of blocks up to the next whole number (6 blocks of 500,000 devices).

- For an Advantage account, the SBCTPS is 30 (6 device blocks * 5 BCTPS).
- For an Essential account, the SBCTPS is 6 (6 device blocks * 1 BCTPS).
- For an Lite account, the SBCTPS is 6 (6 device blocks * 1 BCTPS).

SBCTPS lower limit. Scaled-Base Contracted TPS never falls below the original Base Contracted TPS for the enterprise.

Purchasing Additional TPS

A service provider can offer additional units of TPS to an enterprise through the Marketplace API TPS Service add-on. One TPS unit contains 5 TPS. For more information, see API TPS Service.

This add-on provides two types of additional TPS:

- Incremental Contracted TPS
- Overage TPS

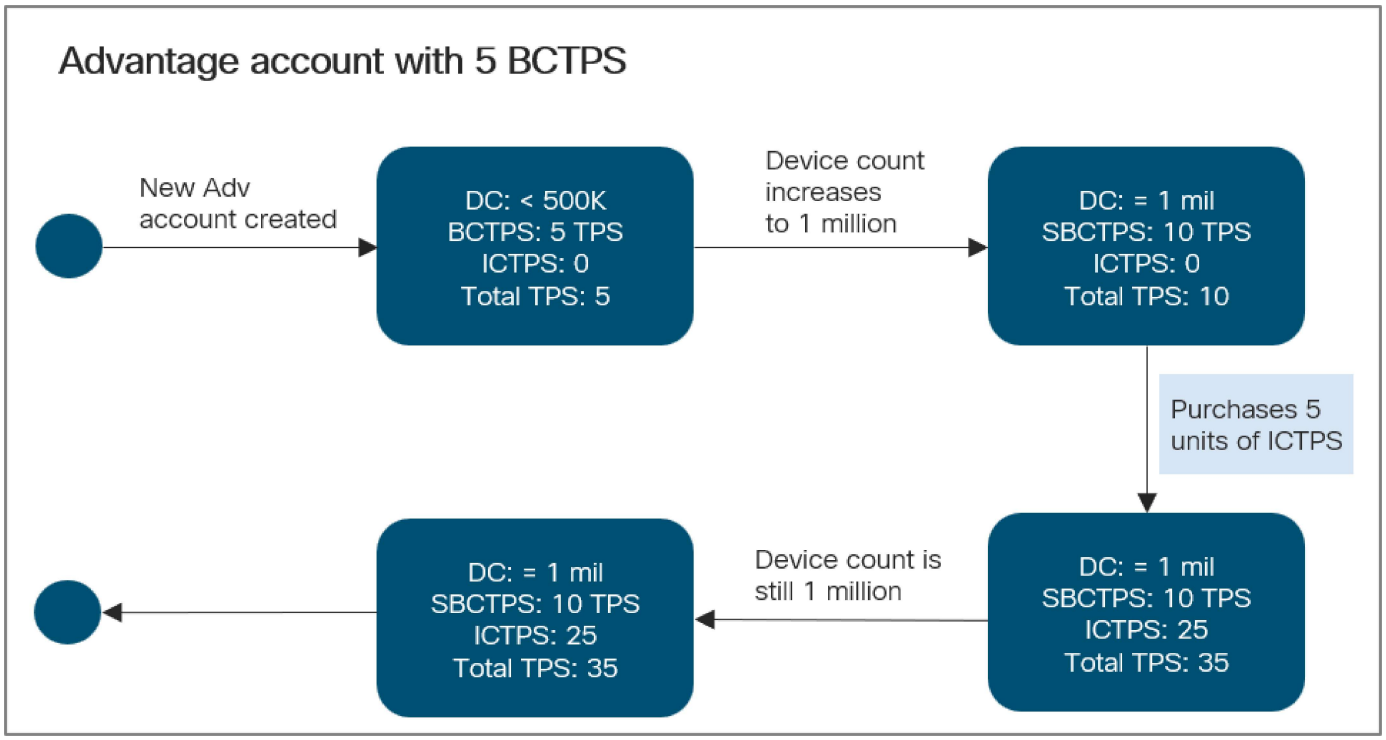
Incremental Contracted TPS

With Incremental Contracted TPS (ICTPS), you pay a monthly fee for additional TPS whether you use it or not. If you exceed the Total Contracted TPS limit (Base/Scaled-Base Contracted TPS + Contracted TPS + Incremental Contracted TPS) and did not opt in for Overage TPS, Control Center throttles the API call and returns an error message. You will need to run the API again later.

Example

An enterprise with a device count of 1 million and a Scaled-Base Contracted TPS of 10, purchases 5 units (25 TPS) of Incremental Contracted TPS. As a result, the Total Contracted TPS for the enterprise

increases from 10 to 35. When the enterprise uses more than 10 TPS, Control Center applies the Incremental Contracted TPS. Control Center throttles all API requests when TPS usage exceeds 35 TPS.



Overage TPS

With Overage TPS (OPTPS), you pay for additional TPS only when you use it. You can opt-in to Overage TPS without purchasing Incremental Contracted TPS, making it a flexible option for those who do not want to pay a monthly fee.

Control Center calculates Overage TPS usage based on the peak TPS level that the enterprise reaches during the billing cycle.

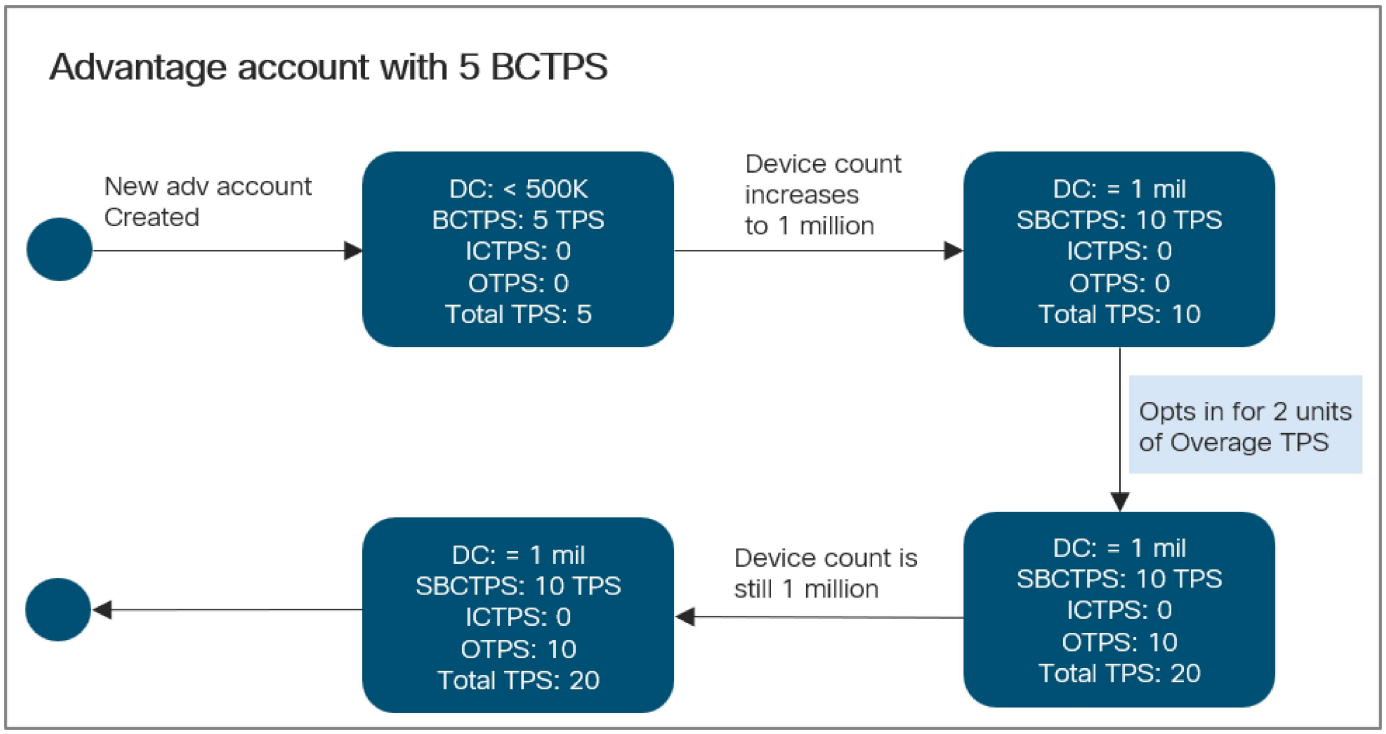
If the enterprise exceeds the Total Contracted TPS limit, the service provider applies the Overage TPS and charges the enterprise accordingly.

If the enterprse exceeds the Total Available TPS (Base/Scaled-Base Contracted TPS + Incremental Contracted TPS + Overage TPS), Control Center throttles the API call and returns an error message. You will need to run the API again later.

Example

An enterprise with a device count of 1 million and a Scaled-Base Contracted TPS of 10, opts in for 2 units (10 TPS) of Overage TPS. As a result, the Total Contracted TPS for the enteprise increases from 10 to 20.

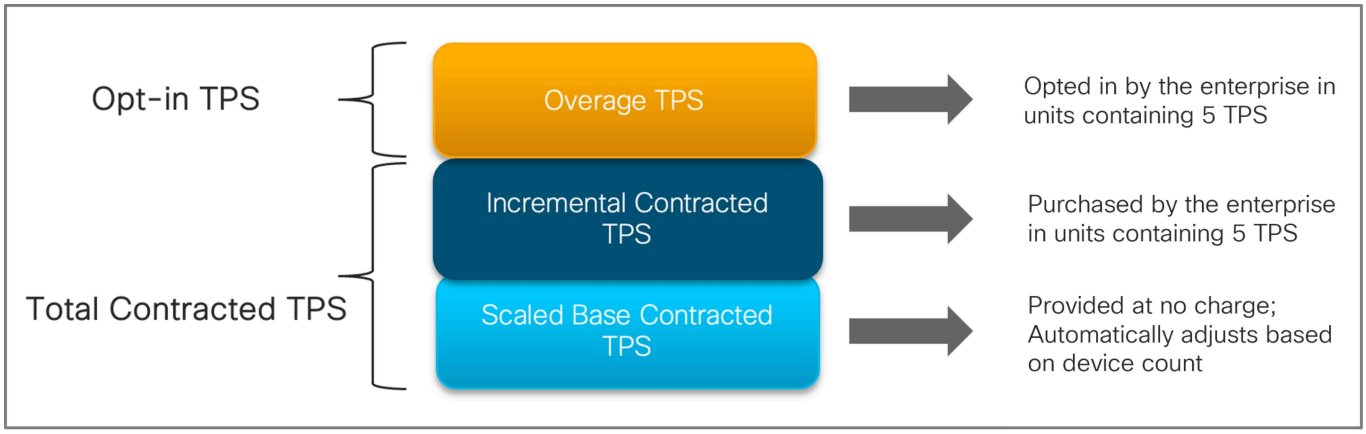
The service provider charges for Overage TPS when TPS usage exceeds 10 TPS. Control Center throttles all API requests when the TPS usage exceeds 20 TPS.



TPS Increments. Control Center offers both Incremental Contracted TPS and Overage TPS in units of 5, with a maximum combined total of 200 units. This excludes Scaled-Base Contracted TPS

Total Available TPS

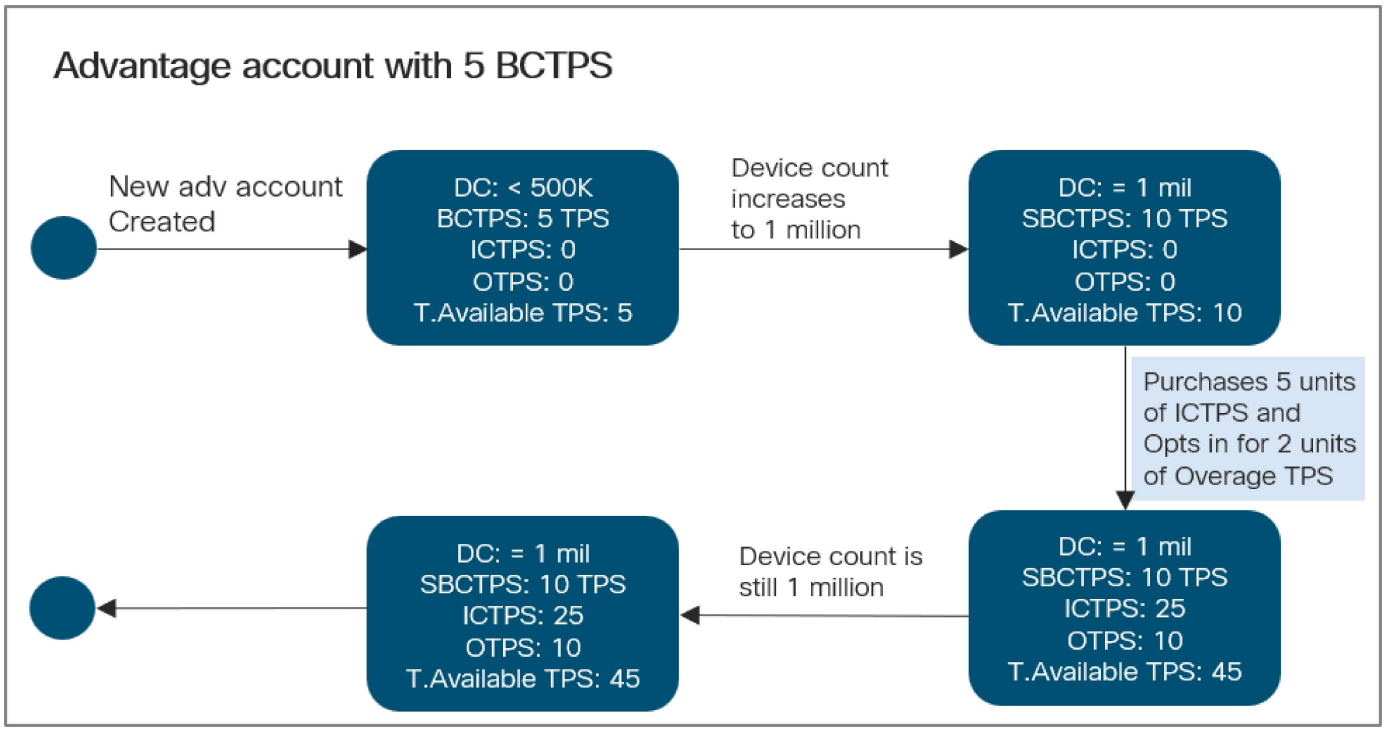
Total Available TPS is the amount of TPS that an enterprise can use before Control Center starts rejecting API requests. Control Center calculates it as: Contracted TPS (Base/Scaled-Base Contracted TPS + Incremental Contracted TPS) + Overage TPS.



Example

An enterprise with a device count of 1 million and a Scaled-Base Contracted TPS of 10, purchases 5 units (25 TPS) of Incremental Contracted TPS and opts in for 2 units (10 TPS) of Overage TPS. As a result,

the Total Available TPS for the enteprise increases from 10 to 45. When the enterprise uses more than 10 TPS, Control Center applies the Incremental Contracted TPS. The service provider charges for Overage TPS when TPS usage exceeds 35 TPS. Control Center throttles all API requests when the TPS usage exceeds 45 TPS.



TPS Limit for 'Write' Operations. Control Center caps APIs that trigger the 'write' operations towards the service provider's core network (e.g., changes to SIM state, Communication Profile changes) at either your maximum Total Available TPS (Total Contracted TPS + Overage TPS) or 25 TPS, whichever is lower.

Viewing API Usage

Control Center provides API Usage Dashboards that give service providers visibility into REST and SOAP API usage for their own account and their enterprise accounts. The API Usage Overview dashboard provides an at-a-glance overview of your API usage by transaction metrics. See API Usage Overview Dashboard. The API Usage Details Dashboard provides a graphical view of your API usage. See API Usage Details Dashboard.

If you offer the API TPS Service add-on to your enterprises, additional TPS-related information appears on the dashboards.

TPS Limits

To effectively manage network traffic, Control Center throttles (rejects) successfully authenticated API requests after an enterprise reaches certain TPS limits. These limits fall into two categories: soft limits

and hard limits.

If you opt-in for Overage TPS, the soft limit is your Total Contracted TPS (Scaled-Base Contracted TPS + Incremental Contracted TPS). At this limit, Control Center continues to accept API requests and charges you for them accordingly.

The hard limit is your Total Available TPS (Total Contracted TPS + Overage TPS). At this limit, Control Center throttles the API requests and provides the following response codes:

API Type	Return Code	Message
REST	429	Rate Limit Exceeded
SOAP	500	License has exceeded the rate limit for API calls

Be aware that this limit includes API calls made by both the account and the account's customers. For example, if an account has two API users and they, along with five of their customers, made simultaneous API calls, Control Center would attribute a total of 7 calls per second to the account.

The limit also applies to the combination of REST and SOAP API calls. For example, suppose an account made 1 REST API call and 1 SOAP API call simultaneously and, at the same time, one of the account's customers made a single REST API call. Control Center would attribute 3 calls per second to the account.

General TPS Policies

As we continue to refine the platform for speed and reliability, Cisco may update the calls per second limit from time to time. These updates may occur without advance notice, however we will document them in the Release Notes.
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Be aware that this limit includes API calls made by both the account and the account's customers. For example, if an account has two API users and they, along with five of their customers, made simultaneous API calls, Control Center would attribute a total of 7 calls per second to the account. The initial API calls would succeed, but after the limit is hit, the rest of the API calls would return one of the messages shown above.

The limit also applies to the combination of REST and SOAP API calls. For example, suppose an account made 1 REST API call and 1 SOAP API call simultaneously and, at the same time, one of the account's customers made a single REST API call. Control Center would attribute 3 calls per second to the account.

Policy: Limit calls to 1 per second for Essential accounts and 5 per second for Advantage accounts.

Best Practices

The following tips can help you avoid throttling or other usage restrictions that may occur when your API use is excessive.

Add Rate Limiting Code

The best way to avoid exceeding TPS limits is to create single-threaded code. However, you may have a scenario where that's not possible. For example:

- Multiple API users are making calls.
- You need to use both REST and SOAP functions.
- You have a high volume of calls requiring multi-threaded processes.

In these cases, we recommend creating a rate limiting routine that monitors and regulates your API calls. For instance, if your account receives numerous error messages when processing API calls, you might introduce a wait time between API requests. You can increase the wait time until the errors stop.

Pseudo Code Example. The following pseudo code illustrates how rate limiting code might work.

```
// DISCLAIMER: THIS EXAMPLE IS PROVIDED FOR ILLUSTRATION PURPOSES ONLY,
// AND WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED

WHILE ( <<list is not empty>> )
{
    // API_call attempts to perform an action to update the DEVICE status
    // it returns the API response code
    response = API_call( <<list element>> )

    // API call returns the "limit exceeded" error, increase loop delay
    IF ( response == ERROR_LIMIT_EXCEEDED )
    {
        delay += ADJUST_DELAY_INCR
    }
    ELSE
    {   // ensure delay never goes below 0
        IF ( delay > ADJUST_DELAY_DECR )
       {
           delay -= ADJUST_DELAY_DECR
       }
       ELSE
       {
           delay = 0
       }
    }

    sleep( delay )
}
```

Adjust API Request Frequency

When tracking device usage with APIs, it is important to align your API requests with the frequency of your service provider updates.

For example, if your service provider updates their usage information once every 6 hours, there's no need to request the information via API more often than once every 6 hours. Also, there is often a delay

between the time when usage data is available and when Control Center receives that usage data from the service provider. In some cases, the delay can be as long as 24 hours.

Use Caching

Cisco encourages local caching. For example, instead of calling an API on every page load, create a simple routine to check the cache once a day.

Use Filters to Limit API Calls

Avoid unnecessary API requests by checking for certain conditions before you execute a program. For example:

- If a device has been deactivated, do not check for usage.
- If an account has not requested usage on a device, do not check every hour.

Lengthen Times Between Searches

If you receive API responses that are empty or unchanged, increase your query time. For example, consider running your queries once a day rather than every two hours.

FTP Reports. For enterprises that require access to large amounts of data on a daily basis, FTP reports may be a better solution than APIs. See FTP Reports for more information.

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